

Levi Neuwirth

RESEARCH ENGINEER — FRONTIER AI EVALUATION & ASSURANCE

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Research engineer designing frontier-model evaluations and reproducible Python pipelines. Combines evaluator calibration and agent failure analysis with systems engineering and rigorous validation of automated verifiers.

RESEARCH & ENGINEERING EXPERIENCE

Cambridge AI Safety Hub — *MARS V Fellow*

July 2026 – Present · Cambridge, United Kingdom

VerInf: zero-knowledge LLM inference research with James Petrie (Future of Life Institute).

- Built and upstreamed a Python workload profiler, versioned manifests, execution DAGs and cost models; validated workload accounting against an archived Llama-4 Maverick run and calibrated on NVIDIA B200 hardware.
- Integrated a collaborator's expert-weight bridge; fixed enrollment-identity binding, Merkle-index checks and sumcheck-round enforcement in Python and Rust verification paths.
- Implemented targeted causal softmax, SiLU and surprisal repairs with a 10-case GPU regression suite; hardened the Rust-verifier harness to distinguish explicit verdicts from crashes or missing output.

VerInf · Case study

Independent Research Contractor — *Frontier AI labs*

2025 – Present

Lead frontier-model evaluation-environment design and evaluator calibration.

- Build tasks and scoring rubrics in **Harbor**; execute models and debug agent scaffolds and trajectories.
- Identify rubrics that measure something other than the intended task; refine evaluation design before scaling runs. Distinguish model failures from defects in task design, scaffolding and scoring.
- Analyze reasoning, capability, safety and red-team failures. Client identities and specific results are confidential.

Proof Broker — Independent Research 2026 – Present. Built an OCaml broker connecting Lean 4 and Rocq to untrusted solvers and model-generated search through versioned evidence schemas, independent certificate checks and kernel-checked proof terms.

Built reproducible Python workflows with pinned tasks, deterministic baselines and negative controls.

Audited context dependencies in proofs; repaired integer-goal reconstruction and the final normalizer. Offline replay produced kernel-checked proofs from 36/36 retained certificates at four obligations with no added axioms.

Shu Laboratory, Brown University Department of Neurology — *Undergraduate Researcher and Technical Lead*

October 2023 – May 2026 · Providence, RI

- Co-led Deep Sets outcome models developed on an 80M+ discharge cohort; readmission AUROC was 0.750 versus 0.655 for the Charlson index on 3.23M temporally held-out discharges. Deployed the public calculator with a Python/FastAPI backend; manuscript under review at JAMIA.
- Technical lead for NeuroPose across four externally funded projects: Python/TensorFlow inference, validated outputs, persistent job state across crashes, Docker deployment and Rust-backed analysis.

EDUCATION

Brown University — *Sc.B., Mathematics and Computer Science*

August 2022 – May 2026 · Providence, RI

GPA 3.8/4.0.

TECHNICAL SKILLS

Languages: Python, Rust, C, C++, Go.

Evaluation: Harbor, Evaluator calibration, Agent scaffolds, Red-teaming.

ML and data: PyTorch, TensorFlow, NumPy, Pandas.

Verification: Lean 4, Rocq, OCaml, SMT (cvc4, cvc5, Z3).

Platforms (Linux): Git, Docker, SLURM.